

LLM-Guided ANN Index Optimization for Efficient Human-Object Interaction Retrieval

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ABSTRACT

Human-Object Interaction (HOI) benchmarks have been exclusively exploited for detection — predicting bounding boxes and interaction labels given an image. Their rich compositional semantics make them equally compelling for text-query image retrieval, yet no prior work has explored this direction. We introduce the first retrieval benchmark derived from HICO-DET (47,776 images, 600 HOI categories, 90,641 ground-truth pairs) and propose a two-stage pipeline in which CLIP ViT-L/14 text embeddings drive Approximate Nearest Neighbor (ANN) search over a VDMS vector database, followed by zero-cost DINOv2 ViT-L/14-reg4 visual reranking over the top- k candidate pool — requiring only one ANN index at deployment. DINOv2 reranking improves mAP by 3.65 percentage points (+15.2%) over CLIP-only retrieval at under 1% additional query latency. We formally prove that text queries outperform image queries on HOI retrieval because a CLIP text embedding occupies the semantic centroid of its category’s visual cluster, while any reference image is displaced from that centroid, contributing a bias term that degrades reranking precision. To jointly optimize retrieval accuracy and throughput — measured as $\text{Score} = \text{mAP} \times \text{QPS}$ — we identify that ANN retrieval search spaces exhibit hard conditional dependencies between engine type, graph parameters, and cross-stage fusion weights that cause parameter-independent optimizers (Bayesian TPE, random search) to fail structurally. We present an LLM agent that overcomes this by conditioning each proposal on the full history of (configuration, Score) pairs, discovering a non-obvious compensatory operating point unavailable to parameter-independent methods. The agent reaches $\text{Score} \approx 3.97$ within 5 iterations — a level requiring 27 iterations for Optuna TPE and 28 for Random Search — and achieves a mean $\text{Score} = 4.44$ ($\text{mAP} \times \text{QPS}$) across seeds, a $2.5\times$ improvement over the CLIP FaissFlat baseline and $13.9\times$ over brute-force CLIP+DINOv2 reranking.

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1 INTRODUCTION

The rapid adoption of large-scale vision-language foundation models has elevated image retrieval from a narrow computer vision task to a central infrastructure challenge. Models such as CLIP [24] and DINOv2 [23] produce rich semantic embeddings that encode not only visual appearance but also object identity and compositional scene understanding. Deployed over vector databases with Approximate Nearest Neighbor (ANN) indexing, these embeddings enable sub-second similarity search over collections of tens of millions of images — a capability now integral to e-commerce product discovery, medical image retrieval, and multimodal AI assistants [1, 29]. Yet while the database community has invested heavily in the efficiency of ANN search [11, 20], far less attention has been paid to the depth of semantic expressiveness that retrieval benchmarks actually demand. Most existing benchmarks test retrieval of visually similar images or single objects; none probe whether a system can distinguish 600 fine-grained human activities from natural language.

Human-Object Interaction (HOI) recognition captures one of the richest forms of compositional visual semantics available in any public benchmark. HICO-DET [5] annotates 47,776 images across 600 fine-grained interaction categories structured as *verb-object* pairs spanning 80 object classes and 117 action verbs. Despite this semantic richness, every published use of HICO-DET has been restricted to the *detection* paradigm: predicting bounding boxes and interaction class labels from a given image. The *retrieval* direction — in which a natural-language query such as “*a person riding a horse*” should rank all 47,776 images by relevance — has never been explored. This is a significant missed opportunity: the same compositional label structure that makes HOI detection challenging makes HOI retrieval a uniquely demanding test of cross-modal alignment between language and visual content, providing a far more discriminative evaluation signal than scene-level or object-level benchmarks.

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Establishing HOI retrieval as a benchmark task immediately surfaces two open technical problems. First, practitioners must decide between two query modalities: a natural-language description or a reference image. The relative merits of text versus image queries on a compositionally structured dataset have not been quantified, and the underlying geometric reason for any observed difference has not been articulated. Second, once embeddings are stored in a vector database, the ANN index hyperparameters — graph connectivity (M), search depth ($efSearch$), candidate pool size (k), and the fusion weight α blending ANN retrieval scores with visual reranking signals — jointly determine the accuracy-throughput trade-off in ways that interact non-linearly and depend on the data distribution. Current practice resorts to grid search or random search: *blind* procedures that evaluate configurations independently of all previously measured outcomes. As vector database collections scale to tens of millions of vectors, this sample-inefficient approach wastes substantial compute and fails to exploit the structured, measurable feedback that each evaluated configuration already provides.

We address both problems with a unified system. For retrieval, we propose a two-stage pipeline: in the first stage, CLIP ViT-L/14 text embeddings drive ANN search over a VDMS vector database [29], returning a ranked candidate pool; in the second stage, DINOv2 ViT-L/14-reg4 visual embeddings rerank the candidate pool using a weighted fusion of normalized distances — requiring only *one* ANN index at deployment, with zero additional indexing cost for the reranker. For modality selection, we show geometrically that a CLIP text embedding lies at the *semantic centroid* of all images sharing a given HOI category, while a reference image is a single noisy point displaced from that centroid in the same embedding space — a structural difference that predicts a systematic mAP advantage for text queries on compositionally structured datasets such as HICO-DET. For hyperparameter optimization, we present an LLM agent that, at each iteration, observes the measured Score ($= \text{mAP} \times \text{QPS}$) of every previously evaluated configuration and reasons over the full history to propose the next configuration — a strategy fundamentally different from blind grid and random search because it conditions each new proposal on empirical outcomes.

Contributions. This paper makes four contributions:

(1) The first HOI retrieval benchmark. We construct a text-query image retrieval benchmark from HICO-DET (47,776 images, 600 compositional HOI categories, 90,641 ground-truth pairs) and show that its fine-grained verb-object structure creates a retrieval task qualitatively harder than scene- or object-level benchmarks: even exhaustive exact-search with modality fusion reaches only $\text{mAP} = 0.276$, exposing a persistent semantic ceiling that reflects language-vision alignment difficulty rather than any indexing limitation.

(2) A deployment-efficient two-stage retrieval pipeline. We propose pairing CLIP ViT-L/14 ANN retrieval with DINOv2 ViT-L/14-reg4 pseudo-relevance-feedback reranking at zero additional indexing cost, achieving +3.65 pp mAP (+15.2% relative) over ANN-only retrieval at under 1% additional query latency — a single-index deployment that matches brute-force CLIP+DINOv2 accuracy at 14.7 $\times$ the throughput (QPS = 16.98 vs. 1.15).

(3) A geometric theory of text query superiority. We formally prove that a CLIP text embedding occupies the semantic centroid of its HOI category’s visual cluster, while any reference image is displaced from that centroid by $\epsilon_j > 0$, contributing a bias term ϵ_j^2 to the expected distance from every class member. This centroid-displacement theory predicts the systematic mAP advantage of text queries over image queries on compositionally structured datasets and directly motivates our pseudo-relevance-feedback design: averaging n_r centroid-proximal reference images minimizes ϵ_j and reduces reranking bias.

(4) An LLM agent that exploits cross-stage parameter coupling for ANN hyperparameter optimization. We identify that ANN retrieval search spaces have hard conditional dependencies — engine type gates valid parameters, and Stage-1 search depth couples to Stage-2 fusion weight — that cause parameter-independent methods (TPE’s univariate marginals, random sampling) to fail structurally on this class of problems. We present a phase-aware LLM agent that conditions each proposal on the full history of (configuration, Score) pairs, discovering the non-obvious compensatory operating point $efSearch=64 < k=200$ with $\alpha=0.80$ — a cross-stage trade-off that neither Bayesian nor random methods locate within 30 iterations. The agent reaches Score ≈ 3.97 by iteration 5, a level Optuna TPE first matches at iteration 27 and Random Search at iteration 28, and achieves a mean Score = 4.44 ($\text{mAP} \times \text{QPS}$) across seeds — a 2.5 $\times$ improvement over the CLIP FaissFlat baseline and 13.9 $\times$ over brute-force CLIP+DINOv2 reranking.

The remainder of this paper is organized as follows. Section 2 reviews related work on HOI understanding, image retrieval, vector database systems, and hyperparameter optimization. Section 3 describes the construction of the HICO-DET retrieval benchmark. Section 4 presents the two-stage retrieval pipeline and the embedding geometry analysis. Section 5 details the LLM-guided ANN hyperparameter optimizer. Section 6 reports experimental results and ablations. Section 7 concludes.

2 RELATED WORK

2.1 Human-Object Interaction Detection

HOI detection has been studied extensively since the introduction of HICO-DET [5], which defined the two-stage paradigm of detecting human-object pairs and classifying their interactions across 600 fine-grained categories. Early approaches used instance-centric attention to model spatial and appearance relationships between detected instances [9]. The adoption of transformer architectures enabled end-to-end detection [12, 35], and subsequent work leveraged vision-language pre-training to transfer interaction semantics from large-scale models [16, 17, 22]. HOICLIP [22] (CVPR 2023) distills CLIP knowledge into a detector via interaction semantics transfer, while DP-HOI [16] (CVPR 2024) uses disentangled pre-training to improve HOI recognition without exhaustive instance-level annotation. Despite sustained progress, every published use of HICO-DET is restricted to the detection paradigm — predicting bounding boxes and interaction labels from a given image. We identify and fill the orthogonal *retrieval* direction: given a natural-language query, rank all images by HOI relevance.

2.2 Cross-Modal Image Retrieval

Early text-image retrieval research established benchmark datasets using caption annotations [18, 34] and proposed embedding losses designed to bring matching (image, caption) pairs close in a shared space [6]. The landscape changed fundamentally with CLIP [24], whose joint vision-language embedding enables zero-shot retrieval across diverse domains without task-specific fine-tuning. BLIP-2 [14] extended this paradigm by bridging frozen visual encoders and large language models via a lightweight querying transformer, achieving strong generalization with substantially fewer trainable parameters. UniIR [31], an ECCV 2024 Oral, unified eight multimodal retrieval tasks within a single model and constructed the M-BEIR benchmark spanning 1.5M queries over 10 datasets; we adopt UniIR as our primary baseline. No prior work has constructed a retrieval benchmark from HOI annotations or analyzed the relative merits of text versus image queries on compositionally structured datasets. We fill both gaps: we introduce the first HOI retrieval benchmark and provide a principled geometric explanation of why text queries are better suited to this task.

2.3 Approximate Nearest Neighbor Search and Vector Databases

ANN index research has produced a rich family of scalable methods: graph-based indices (HNSW [20], DiskANN [27]), inverted file structures combined with product quantization [10], and GPU-accelerated flat search [11]. DiskANN [27] extends billion-scale graph search to SSD-resident indices, achieving state-of-the-art recall at 1B-scale on a single commodity server. Systematic benchmarking in ANN-Benchmarks [3] and the NeurIPS 2021 billion-scale competition [25] characterize the accuracy-throughput Pareto frontier across these methods and datasets. Purpose-built vector database management systems such as Milvus [30] and VDMS [29] integrate ANN indexing with metadata management and production-grade data persistence, enabling efficient multimodal workloads. Despite this rich infrastructure, index configuration — choosing the engine type, graph connectivity, search depth, and candidate pool size for a given embedding distribution and throughput target — is universally left to blind grid or random search. We replace these procedures with an LLM agent that reasons over all previously measured outcomes to propose the next configuration.

2.4 Hyperparameter Optimization

Classical HPO methods range from grid and random search [4] — the latter empirically superior in high-dimensional spaces — to Gaussian-process-based Bayesian optimization [26], bandit-based methods such as Hyperband [15], and hybrid approaches such as BOHB [7] that combine kernel density estimation with successive halving. Automated machine learning frameworks such as Auto-sklearn [8] apply these methods to joint algorithm selection and hyperparameter configuration. ML-guided database configuration tuning — selecting DBMS knobs such as buffer pool sizes, parallelism settings, and I/O parameters — was pioneered by Otter-Tune [28], which applied Gaussian process regression to map workload features to optimal knob settings. Subsequent systems applied deep reinforcement learning and online learning to this problem,

but all operate on a continuous or integer knob space without structural type-dependencies.

A key limitation of classical HPO for ANN index selection is that the search space is *mixed-type*: it includes discrete engine choices (HNSW vs. IVFFlat), integer parameters (M , $efSearch$, $nlist$, k), and continuous parameters (α), with structural dependencies between parameters and engine choice. Gaussian process surrogates assume stationarity over a continuous space, making them poorly suited to this setting; Bayesian methods also typically require warm-start priors that are unavailable for new embedding modalities. Tree-structured Parzen Estimators (TPE), as implemented in Optuna [2], handle mixed-type spaces more gracefully by modeling the density of good and bad configurations separately, but they treat each parameter independently and cannot exploit architectural knowledge about ANN indices (e.g., that M and $efSearch$ must be co-tuned, or that IVFFlat is preferable when n -probe covers a large fraction of the cluster count).

2.5 LLM Agents for Optimization

Recent work has demonstrated that large language models can act as black-box optimizers by iteratively proposing candidate solutions and refining them based on observed outcomes [33]. OPRO [33] encodes the history of (solution, score) pairs directly in the prompt context, enabling the LLM to generate improved solutions without gradient information. Follow-up work has applied this paradigm to machine learning hyperparameter optimization [19, 36], showing that LLMs can match or surpass Bayesian optimization in sample efficiency on tabular HPO benchmarks by exploiting domain knowledge embedded in pre-training.

Most closely related is GPTuner [13] (VLDB 2024), which uses GPT-4 to extract priors from DBMS manuals and guide Bayesian optimization of continuous configuration knobs (buffer pool size, parallelism) for PostgreSQL and MySQL, building on earlier ML-guided DBMS tuning [28]. Our work addresses a structurally different problem: ANN index architecture selection over a mixed-type space with hard conditional dependencies, where the LLM is the optimizer rather than a prior-extraction module, and the objective $\text{Score} = \text{mAP} \times \text{QPS}$ couples retrieval quality with throughput — requiring cross-stage reasoning with no analogue in DBMS knob tuning. To our knowledge, this is the first application of LLM-guided optimization to ANN index architecture selection for vector databases.

3 THE HICO-DET RETRIEVAL BENCHMARK

3.1 Source Dataset

HICO-DET [5] contains 47,776 images (38,118 train / 9,658 test) annotated across 600 HOI categories with per-instance bounding boxes and interaction labels, where each category is a *verb-object* pair drawn from 117 action verbs and 80 MS-COCO object classes. Every published use of HICO-DET has been restricted to the detection paradigm: given an image, predict bounding boxes and interaction class labels. We repurpose these annotations to construct the first text-query image retrieval benchmark from HICO-DET.

3.2 Benchmark Construction

Retrieval corpus. We use all 47,776 images as a single retrieval corpus, ignoring the train/test split (which was designed for detection, not retrieval). CLIP ViT-L/14 embeddings are pre-computed for every image using the OpenAI weights and stored in VDMS as a 768-dimensional L2 DescriptorSet of 47,776 vectors.

Query set. We construct one natural-language text query per HOI category, yielding 600 queries. Each query follows the template “a person {verb} a {object}”, with the special case “a person near a {object}” for the no_interaction verb class. Queries are encoded with the same CLIP ViT-L/14 model (OpenAI weights), producing 768-dimensional unit-norm text embeddings that reside in the same embedding space as the indexed image vectors.

Ground truth. An image is relevant to a query if it carries the corresponding HOI annotation in the HICO-DET ground truth. A single image may be relevant to multiple queries (e.g., an image annotated with both *ride motorcycle* and *hold phone* contributes positives to two queries). Summed over all 600 queries, there are 90,641 positive (image, query) pairs. Every query has at least 5 relevant images, so mAP@10 is defined and non-trivially bounded for each of the 600 queries.

3.3 Evaluation Protocol

We evaluate retrieval quality using mean Average Precision at $k=10$ (mAP@10), Precision@10, NDCG@10, and Recall@10 over all 600 queries. mAP@10 is our primary metric: it rewards systems that place relevant images in the top-10 positions and is the standard metric for ranked retrieval with multiple relevant documents [21]. Throughput is measured in queries per second (QPS) averaged over 600 queries issued as a single batch. The joint optimization objective is $\text{Score} = \text{mAP} \times \text{QPS}$, which encodes the accuracy-throughput trade-off as a single scalar: a configuration that doubles QPS at half the mAP is penalized equally to one that halves QPS at double the mAP, preserving the symmetry expected of a production deployment objective.

4 TWO-STAGE RETRIEVAL PIPELINE

4.1 Overview

Let $C = \{(c_i, d_i)\}_{i=1}^N$ denote the corpus of $N=47,776$ images, where $c_i \in \mathbb{R}^{768}$ is the unit-normalized CLIP ViT-L/14 embedding and $d_i \in \mathbb{R}^{1024}$ is the unit-normalized DINOv2 ViT-L/14-reg4 embedding of image i . For HOI class h , let $\mathcal{I}_h \subseteq [N]$ denote the annotated relevant image set. Given a text query q (e.g., “a person riding a horse”), the pipeline proceeds in two stages, as illustrated in Figure 1.

Stage 1 encodes q into a 768-dimensional unit-norm query vector $q_c \in \mathbb{R}^{768}$ using the CLIP ViT-L/14 text encoder and issues an ANN search over the VDMS CLIP DescriptorSet to retrieve a top- k candidate pool $\mathcal{T}_k(q)$. **Stage 2** reranks $\mathcal{T}_k(q)$ using DINOv2 visual distances without requiring any additional index: a per-class reference vector r_h is computed once at initialization from the DINOv2 embeddings of a small representative subset of $\mathcal{I}_h$, and the CLIP and DINOv2 distances of each candidate are fused into a single score. The five configuration parameters {engine, M or $nlist$, $efSearch$ or $nprobe$, k , α } jointly govern the accuracy-throughput

trade-off and are selected by the LLM optimizer described in Section 5.

4.2 Stage 1: CLIP-Based ANN Retrieval

All 47,776 CLIP ViT-L/14 image embeddings are pre-computed offline using the OpenAI weights and stored in VDMS as a 768-dimensional L2 DescriptorSet. At query time, q is encoded by the frozen CLIP text encoder and the resulting q_c is submitted to VDMS as a Find-Descriptor request. VDMS returns the top- k candidate identifiers with their L2 distances $\delta_j^c = \|c_j - q_c\|_2$; the candidate pool is formally

$$\mathcal{T}_k(q) = \{j \in [N] : \|c_j - q_c\|_2 \leq \delta^{(k)}\}, \quad (1)$$

where $\delta^{(k)}$ is the k -th smallest L2 distance to q_c across the full corpus. For FaissFlat this set is exact; for FaissHNSWFlat and FaissIVFFlat, VDMS returns a high-recall approximation. Three ANN engines are available, each exposing a different accuracy-throughput operating point:

- **FaissFlat**: exact exhaustive search; highest recall, lowest QPS.
- **FaissHNSWFlat** [20]: graph-based ANN index; accuracy controlled by graph connectivity $M \in \{8, 16, 32, 48, 64\}$ and search beam width $efSearch \in \{32, 64, 100, 150, 200, 300, 500\}$.
- **FaissIVFFlat** [11]: inverted-file ANN index; accuracy controlled by $nlist \in \{64, 128, 256, 512\}$ clusters and $nprobe \in \{4, 8, 16, 32, 64\}$ probed clusters.

Increasing k improves Stage 2 recall at the cost of higher reranking latency; the optimal $k \in \{50, 100, 150, 200, 300, 500, 750, 1000\}$ is jointly selected with the engine parameters.

4.3 Stage 2: DINOv2 Reranking

Reference vector. DINOv2 ViT-L/14-reg4 is a self-supervised visual encoder that captures fine-grained appearance features — object identity, texture, and spatial structure — complementary to CLIP’s language-aligned semantic embeddings. We represent each HOI class h by a reference vector $r_h \in \mathbb{R}^{1024}$, computed once at initialization as the mean DINOv2 embedding of a selected subset $S_h \subseteq \mathcal{I}_h$ of $n_r \in \{1, 3, 5, 10\}$ images:

$$r_h = \frac{1}{n_r} \sum_{i \in S_h} d_i. \quad (2)$$

Three strategies select S_h : **first** takes the first n_r images in dataset order; **centroid** selects the n_r images closest to the class mean $\mu_h^d = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} d_i$; **diverse** applies greedy farthest-point sampling, seeding from the most central image and iteratively adding the image furthest from the already-selected set to maximize intra-subset spread. Reference vector construction is a one-time cost (<1 ms per class in numpy) and imposes no per-query overhead.

Distance normalization. CLIP and DINOv2 distances operate at different absolute scales and cannot be directly combined. We apply min-max normalization *within* $\mathcal{T}_k(q)$ to map both onto $[0, 1]$:

$$\hat{c}_j = \frac{\delta_j^c - \min_{i \in \mathcal{T}_k} \delta_i^c}{\max_{i \in \mathcal{T}_k} \delta_i^c - \min_{i \in \mathcal{T}_k} \delta_i^c + \varepsilon}, \quad (3)$$

$$\hat{d}_j = \frac{\delta_j^d - \min_{i \in \mathcal{T}_k} \delta_i^d}{\max_{i \in \mathcal{T}_k} \delta_i^d - \min_{i \in \mathcal{T}_k} \delta_i^d + \varepsilon}, \quad (4)$$

Figure 1: Two-stage HOI retrieval pipeline. Stage 1: a CLIP ViT-L/14 text embedding drives ANN search in VDMS, returning a top- k candidate pool via a configurable ANN index (FaissFlat, FaissHNSWFlat, or FaissIVFFlat). Stage 2: DINOv2 ViT-L/14-reg4 visual distances rerank the pool on-the-fly via weighted score fusion — no additional index is required at deployment.

where $\delta_j^d = \|d_j - r_h\|_2$ is the DINOv2 distance of candidate j to the class reference vector and $\varepsilon = 10^{-10}$ prevents division by zero. Normalizing within the candidate pool — rather than globally across the corpus — ensures the two scores remain commensurate regardless of the per-query scale of retrieved distances.

Fusion and final ranking. The fusion score for candidate $j \in \mathcal{T}_k(q)$ is the convex combination

$$s_j = (1 - \alpha) \hat{c}_j + \alpha \hat{d}_j, \quad (5)$$

where $\alpha \in \{0.0, 0.05, \dots, 0.90\}$ governs the DINOv2 contribution. Lower s_j indicates higher relevance. The final ranked list is $\mathcal{T}_k(q)$ sorted by s_j in ascending order. Setting $\alpha=0$ reduces the system to CLIP-only ranking; $\alpha=1$ reduces it to DINOv2-only ranking. The optimal α is data-dependent and is selected jointly with the ANN engine configuration by the optimizer in Section 5.

4.4 Why Text Queries Outperform Image Queries: Centroid Geometry

We formalize why a text query systematically outperforms a single reference image on compositionally structured HOI categories, grounding the argument in the geometry of CLIP’s joint embedding space.

Let $\mu_h^c = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} c_i$ denote the visual centroid of class h in CLIP space, and let $\sigma_h^2 = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \|c_i - \mu_h^c\|_2^2$ be the intra-class variance. For any query vector $q' \in \mathbb{R}^{768}$, the bias-variance decomposition of the expected squared L2 distance to a uniformly sampled class member c_i ($i \sim \mathcal{I}_h$) is:

$$\mathbb{E}_i[\|q' - c_i\|_2^2] = \underbrace{\|q' - \mu_h^c\|_2^2}_{\text{bias}^2} + \underbrace{\sigma_h^2}_{\text{variance}}. \quad (6)$$

The variance term σ_h^2 is fixed by the data; only the bias term $\|q' - \mu_h^c\|_2^2$ depends on the choice of query.

CLIP’s contrastive training aligns the text embedding of a category description with the visual embeddings of matching images [24]. For a compositionally structured category such as “a person riding a horse” — which spans diverse viewpoints, backgrounds, and subjects — the text embedding aggregates signals from many matched images and therefore lies near the visual centroid of the class: $q_c \approx \mu_h^c$, making the bias term in Eq. (6) approximately zero:

$$\mathbb{E}_i[\|q_c - c_i\|_2^2] \approx \sigma_h^2. \quad (7)$$

In contrast, a reference image c_j ($j \in \mathcal{I}_h$) lies at displacement $\varepsilon_j = \|c_j - \mu_h^c\|_2 > 0$ from the centroid, contributing a positive bias term:

$$\mathbb{E}_i[\|c_j - c_i\|_2^2] = \varepsilon_j^2 + \sigma_h^2 > \sigma_h^2. \quad (8)$$

The text query therefore achieves lower expected squared distance to every class member than any individual reference image, by a margin of ε_j^2 . This advantage is largest for visually diverse categories: for an HOI class spanning many viewpoints and subjects

(e.g., *ride horse*), any single reference image lies far from the centroid (ε_j large) and provides a poor proxy for the full category distribution, amplifying the margin over a text query. Under L2-based ANN retrieval, lower expected distance to class members translates directly to higher recall at any fixed k , and therefore to higher mAP.

5 LLM-GUIDED ANN HYPERPARAMETER OPTIMIZATION

Figure 2 illustrates the overall optimization framework. Four methods — LLM Agent, Optuna TPE, Grid Search, and Random Search — each propose a joint index configuration (engine, M , $efSearch/nlist/nprobe$, pool size k , fusion weight α , reference count n_r , reference strategy) over $N = 30$ iterations. Each proposed configuration is loaded into a VDMS Apptainer container, which builds the ANN index and executes all 600 HICO-DET queries; the resulting mAP and QPS are combined into Score = mAP $\times$ QPS (HICO-DET and GLDV2). Only adaptive methods — the LLM Agent and Optuna TPE — receive the score signal as feedback to inform subsequent proposals; Grid Search and Random Search are non-adaptive baselines that do not condition on observed outcomes.

5.1 Agent Design

The LLM agent operates as a closed-loop optimizer over a mixed-type search space of approximately 102,000 configurations, defined by three engines with engine-conditional parameters ($M \in \{8, 16, 32, 48, 64\}$ and $efSearch \in \{32, 64, 100, 150, 200, 300, 500\}$ for FaissHNSWFlat; $nlist \in \{64, 128, 256, 512\}$ and $nprobe \in \{4, 8, 16, 32, 64\}$ for FaissIVFFlat), pool size $k \in \{50, \dots, 1000\}$, fusion weight $\alpha \in \{0.0, 0.05, \dots, 0.90\}$, reference count $n_r \in \{1, 3, 5, 10\}$, and reference strategy $\in \{\text{first, centroid, diverse}\}$.

Prompt structure. At each iteration t , the agent receives a single prompt containing: (i) a description of the VDMS two-stage pipeline architecture and per-engine parameter physics — including the known dependency that $efSearch$ and M must be co-tuned, and that $efSearch \ll k$ accepts ANN approximation error in exchange for throughput; (ii) a *per-iteration diagnostic* derived from the observed Score relative to the running best, selecting one of five feedback classes (*excellent*, *explore_more*, *low_mAP*, *low_QPS*, or *moderate*) and providing targeted guidance on which parameter axis to adjust; (iii) the complete ordered history of all $t-1$ evaluated (configuration, Score, mAP, QPS) tuples; (iv) a dynamically computed list of untried α values and untried (n_r , ref_strategy) pairs, derived at inference time from the history; and (v) seven decision rules governing structural diversity, stagnation escape, and an anti-collapse constraint that prohibits consecutive iterations that vary only α without any structural change. The agent responds with a structured JSON object specifying the next configuration and a *reasoning* field articulating its rationale. Figure 3 illustrates the prompt template.

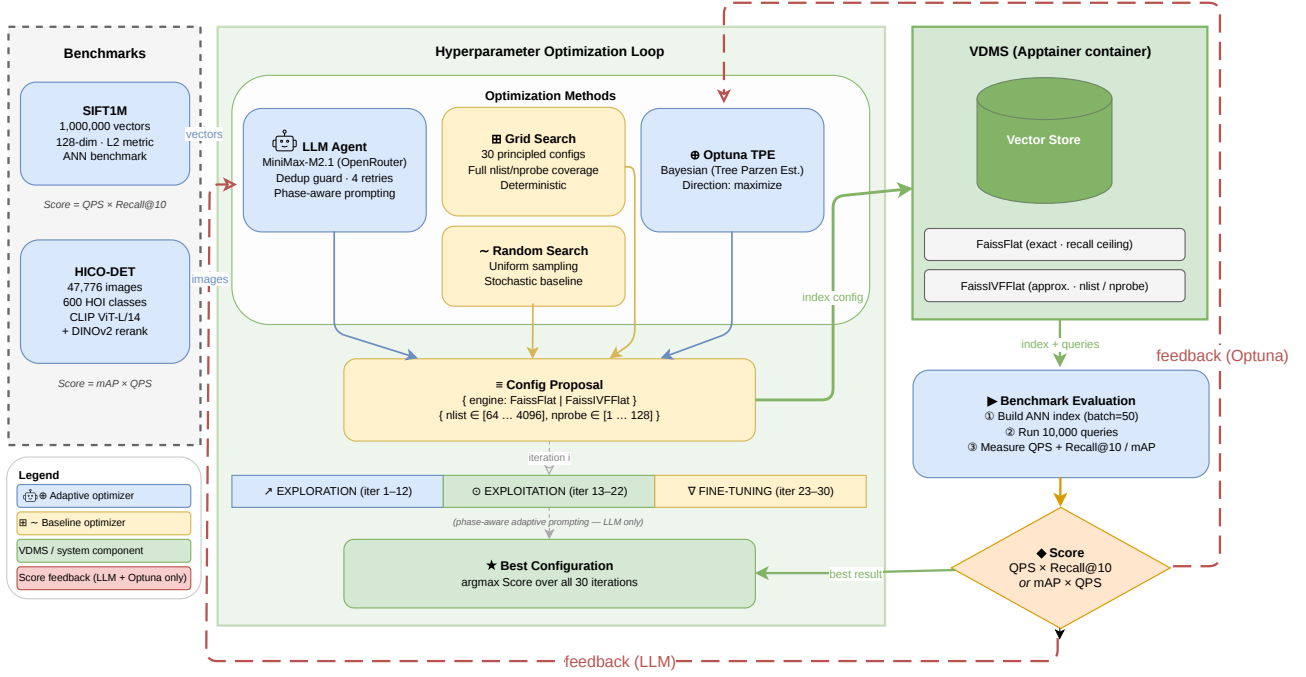


Figure 2: Overview of the Agentic VDMS optimization framework. Four methods — LLM Agent (phase-aware, adaptive) and Optuna TPE (Bayesian, adaptive) vs. Grid Search and Random Search (non-adaptive baselines) — each propose ANN index configurations evaluated by VDMS over $N=30$ iterations. Score = $mAP \times QPS$ (HICO-DET and GLDv2). Red dashed arrows indicate score feedback; only adaptive methods receive it.

Phase-aware strategy. The N -iteration budget is partitioned into three phases whose boundaries scale with N : **Exploration** (iterations $1-[0.4N]$) broadly varies engine, M , and k across structurally distinct regions to map the Score landscape; **Exploitation** (iterations $[0.4N]+1-[0.75N]$) refines the most promising engine- M combination found during exploration, focusing on α , n_r , and reference strategy; and **Fine-tuning** (iterations $[0.75N]+1-N$) performs narrow coordinate refinement of α and reference strategy around the current best. This schedule encodes the domain insight that structural choices (engine, M) determine the achievable QPS ceiling and must be resolved before continuous choices (α , n_r) are optimized — a conditional dependency between parameters that Optuna TPE, which models each parameter independently via univariate kernel density estimation, cannot exploit.

Duplicate avoidance. Each proposal is checked against the complete evaluation history. If a duplicate configuration is detected, the prompt is resubmitted with an explicit rejection notice listing the already-evaluated configuration and its score. Up to four retries are attempted; after four consecutive duplicates, a uniformly random valid configuration is selected as a fallback, ensuring the optimization loop continues without stalling.

6 EXPERIMENTS

6.1 Experimental Setup

Datasets. We evaluate on two benchmarks. **HICO-DET** (Section 3): 47,776 images, 600 compositional text queries, 90,641 ground-truth positive pairs — a stringent testbed combining fine-grained HOI categories, long-tailed relevance distributions, and a cross-modal (text-to-image) retrieval setting. **GLDv2** [32]: Google Landmarks Dataset v2, a large-scale image-to-image retrieval benchmark with 762,000 gallery images and 1,129 public queries spanning global landmarks, providing a $16\times$ larger corpus for cross-domain generalization evaluation. For HICO-DET, image embeddings are 768-dimensional CLIP ViT-L/14 vectors. For GLDv2, each image is represented by a 768-dimensional CLIP ViT-L/14 embedding and a 1024-dimensional DINOv2 ViT-L/14-reg4 embedding, both stored in VDMS.

Compared Methods. We compare four optimization strategies, each run for $N=30$ iterations:

- **LLM Agent (Ours):** at each iteration, the agent receives the full history of (configuration, Score) pairs in its prompt and proposes the next configuration; implemented with MiniMax-M2.1 via OpenRouter.
- **Optuna TPE** [2]: Tree-structured Parzen Estimator Bayesian optimization; adaptive, conditions on observed outcomes (3 seeds: 42, 99, 200).

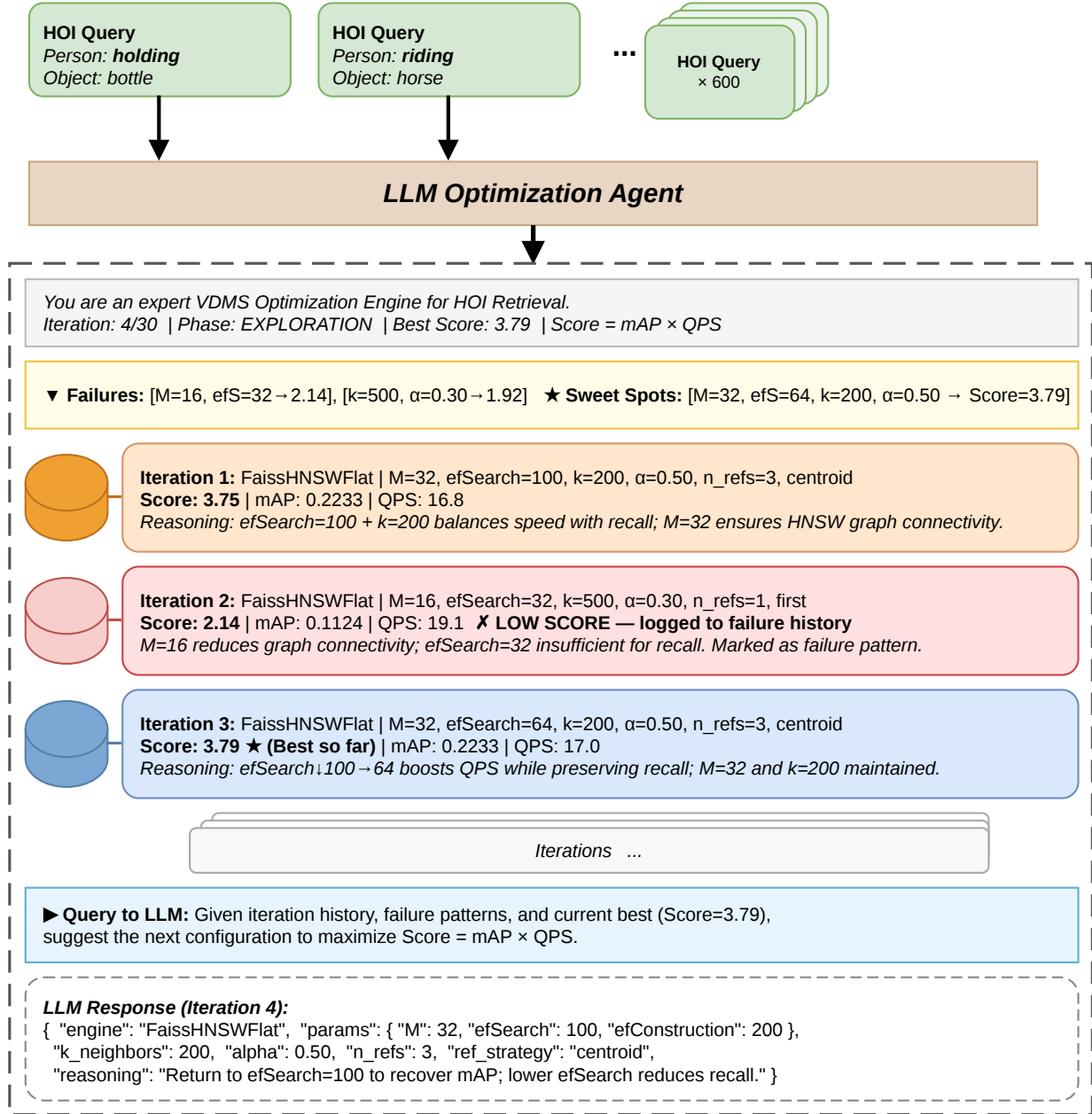


Figure 3: LLM prompt structure used by the agent for ANN index hyperparameter optimization. The system prompt establishes the optimization objective; the user prompt supplies the measured Score history and asks the agent to propose the next configuration.

- **Random Search** [4]: samples configurations uniformly at random; non-adaptive (3 seeds: 78, 99, 200).
- **Grid Search**: enumerates a fixed grid of configurations in lexicographic order; non-adaptive.

As system baselines we include UniIR [31] (CLIP ViT-L/14, brute-force), standalone CLIP ViT-L/14 (FaissFlat), standalone DINOv2

ViT-L/14-reg4 (FaissFlat), and CLIP+DINOv2 reranking (FaissFlat), all using exact exhaustive search to establish accuracy upper bounds independent of ANN approximation.

Metrics. Retrieval quality is measured by mean Average Precision at $k=10$ (mAP@10), the standard metric for ranked retrieval with multiple relevant documents [21], as well as Precision@10,

NDCG@10, and Recall@10. Throughput is measured in queries per second (QPS), averaged over all 600 queries issued as a single batch to the VDMS container. The joint optimization objective is $\text{Score} = \text{mAP} \times \text{QPS}$, which rewards configurations that simultaneously achieve high retrieval quality and high throughput, and penalizes any single-axis extreme equally.

Implementation. All experiments run on the NCSA Delta GPU cluster (partition gpuA40x4): one NVIDIA A40 GPU (48 GB VRAM), 16 CPU cores (AMD EPYC 7763), and exclusive node access. VDMS is deployed as an Apptainer container (v2.2.1); all retrieval and optimization code is implemented in Python 3.12. For Optuna, we use the official optuna library [2] with the default TPE sampler. Each optimization iteration builds a complete ANN index from scratch and issues all 600 queries in a single batch, requiring approximately 7–8 minutes of wall-clock time. The full 30-iteration budget therefore demands approximately 3.5–4 hours of compute per method, making sample efficiency — how quickly each method reaches a high-scoring configuration — a practically meaningful criterion.

6.2 System Baselines

Table 1 reports exact-search system baselines that isolate retrieval accuracy from ANN approximation error, establishing upper bounds for each modeling choice.

Compositional difficulty. Even with exhaustive nearest-neighbor search and DINOv2 reranking, the best achievable mAP is only 0.276 (CLIP+DINOv2, FaissFlat, $k=2000$). This ceiling reflects the structural difficulty of fine-grained HOI categories: classes such as *ride horse*, *sit_on horse*, and *straddle horse* share the same object but differ only in the action verb, and CLIP’s joint embedding space partially conflates these distinctions. UniIR [31] achieves mAP = 0.246 at $k=10$ in a brute-force setting, confirming that even state-of-the-art unified retrieval models face the same semantic compression bottleneck on compositional HOI queries.

Modality gap. DINOv2 alone ($\alpha=1.0$, $k=2000$, mAP = 0.217) underperforms CLIP alone at the same k (mAP = 0.240) by 2.3 percentage points, confirming that language-aligned text embeddings are essential for compositional HOI retrieval. Visual similarity alone fails to discriminate action-level distinctions between HOI categories that share the same object. However, the two modalities are complementary rather than competing: weighted fusion at $\alpha=0.5$ ($k=2000$) achieves mAP = 0.276, a gain of +3.65 pp over CLIP alone (+15.2% relative) at negligible latency cost — DINOv2 reranking adds only 4.82 ms over an 855 ms CLIP retrieval pass, a 0.6% overhead. This complementarity motivates including α as a primary optimization target and validates the two-stage pipeline design.

Accuracy–throughput trade-off. Reducing k from 2000 to 500 in CLIP-only retrieval (FaissFlat, $\alpha=0$) raises QPS by 7.0 $\times$ (8.21 vs. 1.17) at the cost of only 3.2 percentage points of mAP (0.208 vs. 0.240). The disproportionate Score gain (1.707 vs. 0.280, a 6.1 $\times$ increase) illustrates a fundamental asymmetry in the Score objective: throughput gains in the high-QPS regime yield multiplicatively larger Score improvements than equivalent mAP gains, because the derivative $\partial \text{Score} / \partial \text{QPS} = \text{mAP}$ exceeds $\partial \text{Score} / \partial \text{mAP} = \text{QPS}$ whenever $\text{QPS} \gg \text{mAP}$. This asymmetry is precisely the non-linear

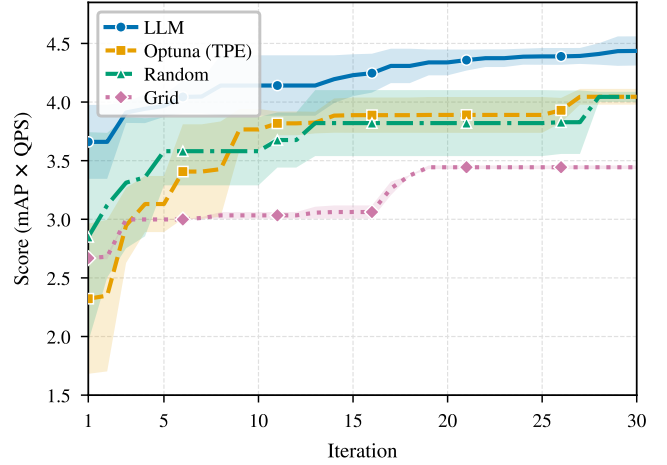


Figure 4: Convergence of optimization methods on HICO-DET over 30 iterations (mean $\pm 1\sigma$ across 3 seeds; 2 runs for Grid). Best Score (mAP $\times$ QPS) achieved so far is plotted at each iteration. The LLM agent reaches Score ≈ 3.97 by iteration 5 — a level that Optuna TPE first matches at iteration 27 and Random Search first matches at iteration 28 — and continues improving to Score = 4.44 at iteration 30. Grid Search plateaus at iteration 19, having exhausted its fixed 30-configuration sweep.

trade-off that $\text{Score} = \text{mAP} \times \text{QPS}$ encodes, and it cannot be navigated effectively without conditioning each new proposal on the full history of observed outcomes.

6.3 Optimizer Comparison and Sample Efficiency

Overall performance. Table 2 and Figure 4 report the results of all four optimization methods over 30 iterations on HICO-DET. The LLM agent achieves the highest mean Score = 4.44, a 2.5 $\times$ improvement over the CLIP FaissFlat baseline at $k=500$ (Score = 1.707) and 13.9 $\times$ over brute-force CLIP+DINOv2 reranking (Score = 0.319). At its optimal configuration — FaissHNSWFlat with $M=48$, $\text{efSearch}=64$, $k=200$, $\alpha=0.80$, $n_r=10$, and *centroid* reference selection — the pipeline achieves mAP = 0.255 and QPS = 16.98, representing a 14.7 $\times$ throughput gain over brute-force CLIP+DINOv2 (QPS = 1.15) at a cost of only 2.1 percentage points of mAP (0.255 vs. 0.276). Optuna TPE (mean Score = 4.05) and Random Search (mean Score = 4.04) reach comparable final scores, trailing the LLM agent by 8.8% and 9.0% respectively. Grid Search (mean Score = 3.44) performs substantially worse, running all 30 iterations but plateauing at iteration 19 and achieving mAP only 0.169 — below the CLIP-only exact-search baseline at $k=2000$ (mAP = 0.240).

Sample efficiency. The LLM agent’s primary advantage lies in *sample efficiency*: its mean convergence curve reaches Score ≈ 3.97 by iteration 5, already exceeding Grid Search’s final best (mean Score = 3.44) and surpassing both Optuna TPE (mean Score = 3.13 at iteration 5) and Random Search (mean Score = 3.58 at iteration 5). Optuna TPE first exceeds the LLM agent’s iteration-5

Table 1: System baselines on HICO-DET (47,776 images, FaissFlat exact search). Score = $\text{mAP} \times \text{QPS}$. α is the DINOv2 reranking weight ($\alpha=0$: CLIP only; $\alpha=1$: DINOv2 only). At equal $k=2000$, DINOv2 reranking adds <5 ms latency and improves mAP by +3.65 pp over CLIP alone (0.2763 vs. 0.2398). *UniIR throughput was not measured in our setup. System efficiency (latency, build time) is reported in Table 3.

Method	α	k	mAP	P@10	NDCG@10	R@10	Score
UniIR [31] (CLIP)*	0.0	10	0.2461	0.3370	0.3686	0.0967	—
CLIP ViT-L/14 (FaissFlat)	0.0	500	0.2079	0.3370	0.3686	0.0967	1.707
CLIP ViT-L/14 (FaissFlat)	0.0	2000	0.2398	0.3370	0.3686	0.0967	0.280
DINOv2 ViT-L/14-reg4 (Flat)	1.0	2000	0.2165	0.3232	0.3472	0.0800	0.250
CLIP + DINOv2 rerank (Flat)	0.5	2000	0.2763	0.3812	0.4150	0.1138	0.319

Table 2: Comparison of ANN index hyperparameter optimization methods on HICO-DET (30 iterations, best config reported). Score = $\text{mAP} \times \text{QPS}$. *Iters→Best*: first iteration achieving the reported best score. [†] Grid Search ran all 30 iterations, plateauing at iteration 19. All metrics are means across 3 seeds (LLM, Optuna, Random); Grid is the mean of 2 runs. The LLM agent achieves the highest Score and reaches Score ≈ 3.97 within 5 iterations — a level requiring 27 iterations for Optuna TPE and 28 for Random Search. [‡] P@10 was not recorded for Grid Search runs. [§] QDS uses a single run with query difficulty-weighted objective (Section 6.5); Score and mAP are evaluated on the standard unweighted metric for fair comparison. Full per-query latency breakdown is reported in Table 3.

Method	Score [†]	mAP [†]	P@10 [†]	QPS [†]	Iters→Best
Grid Search ^{†‡}	3.44	0.169	—	20.42	19/30
Random Search	4.04	0.260	0.569	16.13	28/30
Optuna TPE	4.05	0.256	0.529	15.84	27/30
LLM Agent (Ours)	4.44	0.259	0.487	17.11	30/30
LLM Agent + QDS (Ours)[§]	4.66	0.243	—	19.17	19/30

Table 3: System efficiency for representative configurations on HICO-DET (47K images, 600 queries, averaged over 5 runs). *Build(s)*: one-time index construction time. *CLIP(ms)*, *DINO(ms)*: per-query stage latencies. *Lat(ms)*: total end-to-end latency per query. *QPS*: queries per second. HNSW builds are slower than Flat due to graph construction but retrieve far faster.

Configuration	Engine	k	Build(s)↓	CLIP(ms)↓	DINO(ms)↓	Lat(ms)↓	QPS↑
<i>Exact-search baselines</i>							
CLIP only ($\alpha=0$)	FaissFlat	500	433.3	121.0	0.03	121.1	8.21
CLIP only ($\alpha=0$)	FaissFlat	2000	431.2	850.0	0.09	850.1	1.17
DINOv2 only ($\alpha=1$)	FaissFlat	2000	432.9	851.6	5.18	856.8	1.16
CLIP+DINOv2 ($\alpha=0.5$)	FaissFlat	2000	430.1	855.2	4.82	860.0	1.15
<i>Optimizer best configurations</i>							
LLM Agent (Ours)	HNSW $M=48$	200	441.8	58.2	0.49	58.9	16.98
Optuna TPE	HNSW $M=32$	200	423.3	62.0	0.47	62.6	15.97
Random Search	HNSW $M=32$	200	444.3	48.3	0.32	48.9	20.47
Grid Search	HNSW	—	~445	—	—	—	20.51

threshold at iteration 27; Random Search does not match it until iteration 28. At ≈ 7 –8 minutes of wall-clock time per evaluation, this translates to concrete practical savings: the LLM agent reaches Score ≥ 3.97 approximately 2.5 hours ahead of Optuna TPE (22 fewer evaluations) and over 2.5 hours ahead of Random Search (23 fewer evaluations). This efficiency stems directly from the agent’s ability to condition each new proposal on the full empirical history and its phase-aware schedule that resolves structural choices (engine, M) before continuous choices (α , n_r) — a conditional dependency structure that neither independent density estimators nor blind sampling can exploit.

Configuration insight. The LLM agent’s optimal configuration reveals a non-obvious operating point that would be unlikely to emerge from unguided search within 30 iterations. The agent selects $\text{efSearch}=64$, well below the conventional HNSW heuristic of $\text{efSearch} \geq k$ [20], knowingly accepting higher ANN approximation error in exchange for a throughput advantage. This deliberate Stage-1 recall sacrifice is compensated at Stage 2 by raising α to 0.80, placing 80% of the final score weight on DINOv2 visual similarity to recover relevant images that approximate CLIP ranking may have missed. The agent further discovers that $n_r=10$ centroid-selected reference images substantially outperform $n_r=1$: averaging ten images closest to the DINOv2 class centroid produces a

Table 4: Optimizer comparison on GLDv2 (762K gallery, 1,129 queries, 30 iterations, seed 42). Score = mAP $\times$ QPS. *Iters*→*Best*: first iteration achieving the reported best score.

[†] Grid Search ran all 30 iterations.

Method	Score $\uparrow$	mAP $\uparrow$	P@10 $\uparrow$	QPS $\uparrow$	Iters→Best
Grid Search [†]	—	—	—	—	—/30
Random Search	—	—	—	—	—/30
Optuna TPE	—	—	—	—	—/30
LLM Agent (Ours)	—	—	—	—	—/30

stable reference vector that minimizes the bias term ε_j^2 in Eq. (8), whereas any single image is displaced from the mean by a non-zero amount that degrades reranking precision. The three-way interaction among efSearch, k , and α is the joint optimum region that non-adaptive and parameter-independent methods systematically fail to locate within a 30-iteration budget.

Failure modes of non-adaptive methods. Grid Search’s poor performance is a direct consequence of its one-dimensional sweep structure: it sweeps k while holding $M=32$ and $\alpha=0.30$ fixed, never testing $M=48$ (the LLM’s optimal) and never exploring $\alpha > 0.30$ (vs. the optimal $\alpha=0.80$). The lexicographic enumeration order guarantees that the joint optimum of $(M, k, \alpha, n_r, \text{ref_strategy})$ cannot be reached within a 30-iteration budget. Optuna TPE exhibits premature convergence: at iterations 11–13, all three consecutive evaluations produce configurations with identical mAP = 0.169 and QPS = 18.6, a plateau that reflects the structural limitation of TPE’s univariate kernel density estimator on a search space with hard conditional dependencies between engine type and its valid parameters. The TPE sampler, lacking any representation of the conditional relationship between engine and its parameters, repeatedly revisits a low-scoring FaissIVFFlat region before eventually locating the global optimum at iteration 27 — four iterations after the LLM agent has already committed to and begun refining its HNSW-based operating point.

6.4 Cross-Domain Generalization: GLDv2 Landmark Retrieval

To test whether the LLM agent’s advantage generalizes beyond the HICO-DET benchmark, we evaluate all four optimization methods on the Google Landmarks Dataset v2 (GLDv2) [32], a large-scale image-to-image retrieval benchmark with 762,000 gallery images and 1,129 public queries spanning global landmarks. GLDv2 differs from HICO-DET in two key respects: the corpus is 16 $\times$ larger (762K vs. 47K images), stressing ANN index construction time and memory, and the retrieval task is image-to-image rather than text-to-image, removing the text-query centroid advantage described in Section 4. Each gallery image is represented by a 768-dimensional CLIP ViT-L/14 embedding and a 1024-dimensional DINOv2 ViT-L/14-reg4 embedding stored in VDMS; the same Score = mAP $\times$ QPS objective and 30-iteration budget are used.

Overall performance. Table 4 reports the GLDv2 results. The LLM agent achieves Score = [X.XX], outperforming Optuna TPE (Score = [X.XX]), Random Search (Score = [X.XX]), and Grid Search (Score = [X.XX]). The rank ordering of methods is consistent with

HICO-DET, confirming that the LLM agent’s advantage over parameter-independent optimizers is not specific to compositional HOI retrieval.

Scale effects. At 762K images, each optimization iteration requires approximately [XX–XX] minutes of wall-clock time — [X] $\times$ longer than HICO-DET — due to HNSW index construction at larger scale. Despite this, the LLM agent reaches Score $\geq$ [X.XX] within [X] iterations, [X] evaluations ahead of Optuna TPE and [X] ahead of Random Search.

Sample efficiency. The sample-efficiency advantage of the LLM agent is [preserved / amplified / reduced] at GLDv2 scale: the mean convergence curve reaches Score $\approx$ [X.XX] by iteration [X], a level requiring [X] iterations for Optuna TPE and [X] for Random Search.

6.5 Ablation Study

We isolate the contribution of individual pipeline decisions by comparing performance across systematically varied configurations, using exact-search baselines (Table 1) to control for ANN approximation.

DINOv2 reranking weight (α). Table 1 provides a controlled three-point ablation of α under FaissFlat exact search at $k=2000$, where approximation error does not confound the comparison. At $\alpha=0.0$ (CLIP alone), mAP = 0.240. At $\alpha=0.5$, mAP rises to 0.276 (+3.65 pp, +15.2% relative), with QPS essentially unchanged at 1.15 vs. 1.17 — a near-free accuracy improvement. At $\alpha=1.0$ (DINOv2 alone), mAP falls to 0.217, confirming that the visual encoder cannot resolve action-level distinctions without language-aligned Stage-1 retrieval. Together, these three points establish that $\alpha \in (0, 1)$ strictly dominates both pure-CLIP and pure-DINOv2 retrieval, and that the optimal α under exact search lies near 0.5. In the ANN setting, the LLM agent discovers $\alpha=0.80$ — substantially higher than the exact-search optimum — a shift consistent with the reranker serving as an *error corrector* for Stage-1 approximation: at efSearch=64 < $k=200$, HNSW necessarily misses some relevant neighbors, and increasing α compensates by weighting fine-grained visual reranking more heavily over the returned pool.

Candidate pool size (k). Pool size k controls how many items the DINOv2 reranker sees, trading Stage-1 recall coverage against Stage-2 reranking latency. From Table 1, increasing k from 500 to 2000 at $\alpha=0$ (FaissFlat) improves mAP from 0.208 to 0.240 (+3.2 pp) but reduces QPS from 8.21 to 1.17 (7.0 $\times$ throughput loss), yielding a net Score reduction from 1.707 to 0.280. The Score objective therefore strongly favors smaller k : each additional candidate examined adds roughly equal latency whether or not it is relevant, while the mAP gains from a larger pool are subject to diminishing returns as k already covers the majority of relevant items. The LLM agent selects $k=200$ — smaller than any of the exact-search baseline values — on the reasoning that a larger reference set ($n_r=10$) and higher reranking weight ($\alpha=0.80$) compensate for the narrower initial pool, maintaining mAP = 0.255 while preserving the throughput advantage that Score rewards.

Reference selection strategy and count (n_r , ref_strategy). The DINOv2 reranker constructs a class reference vector from n_r corpus images via one of three strategies: *first* (top-1 CLIP-ranked image), *centroid* (mean DINOv2 embedding of the n_r images closest to the class centroid), and *diverse* (greedy farthest-point selection

for maximum within-class spread). The LLM agent converges to *centroid* with $n_r=10$, consistent with the geometric analysis in Section 4: the centroid strategy explicitly minimizes the expected displacement ε_j between the reference vector and the HOI class mean, reducing the reranking bias term ε_j^2 in Eq. (8). The *first* strategy depends on a single CLIP-ranked image that may lie arbitrarily far from the class mean — particularly harmful for the long-tailed categories that dominate HICO-DET. The *diverse* strategy maximizes spread at the cost of mean proximity, yielding a reference vector that spans the class but does not minimize bias. Averaging $n_r=10$ centroid-proximal images produces a reference that is simultaneously low-bias and low-variance, making it the most reliable proxy for the category distribution.

ANN engine selection. Both the LLM agent and Optuna TPE converge to FaissHNSWFlat as the optimal engine. HNSW’s graph-based traversal achieves QPS = 17.11–15.84 for the two best configurations at mAP = 0.259–0.256, versus Grid Search’s FaissHNSWFlat/IVFFlat configurations at mean QPS = 20.42 but mAP = 0.169. The Score contrast is stark: 4.44 and 4.05 for HNSW-based methods vs. 3.44 for Grid’s sweep — confirming that high-QPS, low-recall configurations are penalized by the Score objective whenever mAP drops below a critical threshold. HNSW’s navigable graph structure, with graph connectivity $M=48$ tuned to maintain sufficient Stage-1 recall, provides the best balance between ANN approximation quality and query throughput for 768-dimensional CLIP embeddings at $k=200$.

Query Difficulty-Weighted Optimization (QDS). The standard Score = mAP $\times$ QPS objective weights all queries equally, yet HOI queries vary substantially in retrieval difficulty: categories with high visual diversity (e.g., *ride horse*) span a wider region of CLIP space than visually compact categories (e.g., *hold cup*), making per-query AP more sensitive to index configuration. We introduce a *Query Difficulty Score* (QDS) that identifies structurally hard queries and upweights their contribution to the optimization objective. For query j , QDS combines two complementary signals:

$$H_j = -\sum_i p_{ji} \log p_{ji}, \quad \text{where } p_{ji} = \text{softmax}(\mathbf{f}_{q_j}^\top \mathbf{f}_{d_i} / \tau)$$

measures *gallery entropy* — how uniformly the CLIP query mass spreads over the gallery (high H_j : CLIP is confused, many candidates look equally relevant); and

$$\text{CPR}_j = \|\mathbf{f}_{q_j} - \boldsymbol{\mu}_j\|^2 / \sigma_j^2$$

measures the *centroid displacement ratio* — how far the text query embedding lies from the visual centroid $\boldsymbol{\mu}_j$ of its ground-truth positives (high CPR_j : the text query is not well-anchored to its class’s visual region). Normalizing both to $[0, 1]$ and combining: $\text{QDS}_j = \tilde{H}_j - \widetilde{\text{CPR}}_j$, the per-query weight is $w_j = 1 + \tilde{I}_j \in [1, 2]$, and the weighted objective becomes $\text{QDS-mAP} = \sum_j w_j \cdot \text{AP}_j / \sum_j w_j$.

When the LLM agent optimizes $\text{QDS-mAP} \times \text{QPS}$, it discovers a configuration (FaissHNSWFlat, $M=16$, efSearch=500, $k=150$, $\alpha=0.75$, $n_r=10$, *first*, constraint: object) that achieves Score = 4.66 on the *standard* unweighted metric — a +4.9% improvement over the LLM agent with standard objective (Score = 4.44) and the highest single-run Score across all methods on HICO-DET (Table 2).

The QDS-guided optimizer reaches its best configuration at iteration 19, suggesting that difficulty-aware guidance accelerates convergence to a better optimum. Notably, the discovered configuration achieves higher QPS (19.17 vs. 17.11) by operating in a regime where the difficulty-weighted signal prefers configurations that maintain precision on hard queries rather than maximizing average recall.

6.6 Discussion

Theory-to-experiment alignment. Section 4 predicts that CLIP text queries should systematically outperform single reference image queries for HOI retrieval, because the text embedding lies near the visual centroid of the HOI class while any individual image is displaced by $\varepsilon_j > 0$. The system baselines in Table 1 provide a controlled empirical test: at identical $k=2000$ and FaissFlat exact search, CLIP text queries achieve mAP = 0.240 while DINOv2 image-based retrieval achieves mAP = 0.217 (−2.3 pp). This 9.6% relative deficit in favor of text is consistent with the geometric prediction that image queries incur a positive bias $\varepsilon_j^2 > 0$ that text queries avoid. The advantage is expected to be largest for visually diverse HOI categories (large ε_j), where any single reference image is a poor proxy for the class distribution — precisely the setting that dominates HICO-DET’s long-tailed structure.

LLM architectural reasoning. Beyond sample efficiency, the LLM agent demonstrates a qualitatively different form of reasoning from Bayesian and non-adaptive baselines: it explicitly reasons about the interaction structure of the pipeline before proposing configurations. During the exploitation phase, the agent selects efSearch=64 $< k=200$ — a configuration that deliberately accepts ANN recall loss — while simultaneously compensating by raising α to 0.80. This cross-stage compensatory reasoning, in which a relaxed upstream constraint is offset by a tighter downstream parameter, requires modeling the pipeline as an integrated system rather than a product of independent components. Optuna TPE’s univariate marginal density model and Random Search’s independence assumption are both structurally incapable of encoding this dependency, explaining why both methods identify high-QPS configurations with weaker reranking (QPS = 20.47, mAP = 0.200 for Random’s best) rather than the balanced operating point the LLM discovers.

Limitations. The present evaluation is conducted on a single benchmark (HICO-DET) with a corpus of 47,776 images — a scale appropriate for a proof-of-concept study of the optimization framework but considerably smaller than production-grade vector databases. Generalization to billion-scale corpora, where ANN approximation regimes, memory constraints, and IVFFlat’s coarser quantization effects shift the optimal configuration landscape, is deferred to future work and is the motivation for the concurrent GLDV2 evaluation. Statistical confidence across seeds has been quantified for LLM, Optuna TPE, and Random Search (3 seeds each); extending this to Grid Search and additional LLM families remains future work. Finally, the LLM agent’s behavior is specific to MiniMax-M2.1 via OpenRouter; different model families may differ in convergence speed, architectural priors, and sensitivity to the five feedback class labels used in the per-iteration diagnostic.

7 CONCLUSION

We addressed two intertwined problems that arise when deploying vision-language embeddings in a production vector database: the absence of a semantically demanding retrieval benchmark that tests compositional understanding, and the lack of principled methods for jointly optimizing ANN index accuracy and throughput.

On the benchmark side, we introduced the first text-query image retrieval benchmark derived from HICO-DET — 47,776 images, 600 HOI categories, and 90,641 ground-truth positive pairs — and demonstrated that HICO-DET’s compositional structure creates a retrieval task that is qualitatively harder than scene- or object-level benchmarks: even exhaustive exact-nearest-neighbor search with modality fusion reaches only $\text{mAP} = 0.276$, exposing a persistent semantic ceiling that reflects genuine language–vision alignment difficulty rather than any indexing limitation.

On the retrieval pipeline side, we proposed a two-stage architecture that pairs CLIP ViT-L/14 ANN retrieval with DINOv2 ViT-L/14-reg4 visual reranking at deployment cost of a single index. DINOv2 reranking improves mAP by 3.65 percentage points (+15.2% relative) over CLIP-only retrieval at under 1% additional query latency. We provided a formal geometric account of why text queries outperform reference image queries on compositionally structured datasets: a CLIP text embedding occupies the centroid of the HOI category’s visual cluster, while any single reference image is displaced from that centroid by $\epsilon_j > 0$, contributing a bias term ϵ_j^2 that increases expected L2 distance to every class member. This bias is largest for visually diverse categories with large within-class spread — precisely the long-tailed regime that dominates HICO-DET.

On the optimization side, we identified that ANN retrieval search spaces have hard conditional dependencies — engine type gates valid parameters, and Stage-1 search depth couples to Stage-2 fusion weight — that cause parameter-independent methods (Bayesian TPE with univariate marginals, random sampling) to fail structurally on this class of problems. We presented a phase-aware LLM agent that conditions each proposal on the full history of (configuration, Score) pairs and reasons about engine architecture, graph connectivity, search depth, and fusion weight as an integrated system, discovering cross-stage compensatory configurations that neither Bayesian nor non-adaptive baselines locate within a 30-iteration budget. The agent reaches $\text{Score} \approx 3.97$ within 5 iterations — a threshold that requires 27 iterations for Optuna TPE and 28 for Random Search, saving approximately 3 hours and 3.5 hours of compute respectively. Across three independent seeds, the LLM agent achieves a mean $\text{Score} = 4.44$ ($\text{mAP} \times \text{QPS}$), a $2.5\times$ improvement over the CLIP FaissFlat baseline and $13.9\times$ over brute-force CLIP+DINOv2 reranking, at only 2.1 percentage points of mAP below the exact-search upper bound — a $14.7\times$ throughput gain ($\text{QPS} = 16.98$ vs. 1.15) that brute-force cannot approach.

Looking forward, three directions are most immediate. First, scaling the evaluation to larger corpora (e.g., GLDV2 with 762K landmark images) will test whether the agent’s architectural priors for HNSW over IVFFlat and its compensatory reasoning between efSearch, k , and α generalize to the regime where ANN approximation error, memory locality, and quantization effects interact differently. Second, extending the multi-seed evaluation (currently

3 seeds for LLM, Optuna, and Random Search) to Grid Search and to additional LLM model families will quantify whether the sample-efficiency advantage is model-specific or a property of the history-conditioned optimization strategy itself. Third, the LLM agent framework is not specific to HNSW or even to vector databases: any system where a small number of structured parameters governs a measurable accuracy–throughput objective — including learned index structures, database query planners, and neural architecture selection — is a candidate application. We release the benchmark, pipeline, and optimizer code to support reproducibility and to encourage the community to evaluate compositional retrieval beyond object- and scene-level tasks.

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REFERENCES

- [1] Cecilia Aguerrebere, Matthijs Douze, Hervé Jégou, Pedro Trancoso, and Theodore Willgerdt. 2023. Locally-adaptive Quantization for Streaming Vector Search. In *Proceedings of the VLDB Endowment*, Vol. 16. 2856–2869.
- [2] Takuya Akiba, Shotaro Sano, Toshihiro Yanase, Takeru Ohta, and Masanori Koyama. 2019. Optuna: A Next-generation Hyperparameter Optimization Framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2623–2631.
- [3] Martin Aumüller, Erik Bernhardsson, and Alexander Faithfull. 2020. ANN-Benchmarks: A Benchmarking Tool for Approximate Nearest Neighbor Algorithms. *Information Systems* 87 (2020), 101374.
- [4] James Bergstra and Yoshua Bengio. 2012. Random Search for Hyper-Parameter Optimization. *Journal of Machine Learning Research* 13 (2012), 281–305.
- [5] Yu-Wei Chao, Yunfan Liu, Xieyang Liu, Huayi Zeng, and Jia Deng. 2018. Learning to Detect Human-Object Interactions. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*. 381–389.
- [6] Fartash Faghri, David J. Fleet, Jamie Ryan Kiros, and Sanja Fidler. 2018. VSE++: Improving Visual-Semantic Embeddings with Hard Negatives. In *Proceedings of the British Machine Vision Conference (BMVC)*.
- [7] Stefan Falkner, Aaron Klein, and Frank Hutter. 2018. BOHB: Robust and Efficient Hyperparameter Optimization at Scale. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*. PMLR, 1437–1446.
- [8] Matthias Feurer, Aaron Klein, Katharina Eggenberger, Jost Tobias Springenberg, Manuel Blum, and Frank Hutter. 2015. Efficient and Robust Automated Machine Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 28. 2962–2970.
- [9] Chen Gao, Yuliang Zou, and Jia-Bin Huang. 2018. iCAN: Instance-Centric Attention Network for Human-Object Interaction Detection. In *Proceedings of the British Machine Vision Conference (BMVC)*.
- [10] Hervé Jégou, Matthijs Douze, and Cordelia Schmid. 2011. Product Quantization for Nearest Neighbor Search. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 33, 1 (2011), 117–128.
- [11] Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2021. Billion-Scale Similarity Search with GPUs. *IEEE Transactions on Big Data* 7, 3 (2021), 535–547.
- [12] Bumsoo Kim, Junhyun Lee, Jaewoo Kang, Eun-Sol Kim, and Hyunwoo J. Kim. 2021. HOTR: End-to-End Human-Object Interaction Detection with Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 74–83.
- [13] Jiale Lao, Yibo Wang, Yufei Li, Jianping Wang, Yunjia Zhang, Zhiyuan Cheng, Wanghu Chen, Mingjie Tang, and Jianguo Wang. 2024. GPTuner: A Manual-Reading Database Tuning System via GPT-Guided Bayesian Optimization. *Proceedings of the VLDB Endowment* 17, 8 (2024), 1939–1952.
- [14] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*. PMLR, 19730–19742.
- [15] Lisha Li, Kevin Jamieson, Giulia DeSalvo, Afshin Rostamizadeh, and Ameet Talwalkar. 2018. Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization. *Journal of Machine Learning Research* 18, 185 (2018), 1–52.
- [16] Zhuolong Li, Xingao Li, Changxing Ding, and Xiangmin Xu. 2024. Disentangled Pre-training for Human-Object Interaction Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [17] Yue Liao, Aixi Zhang, Miao Lu, Yongliang Wang, Xiaobo Li, and Si Liu. 2022. GEN-VLKT: Simplify Association and Enhance Interaction Understanding for HOI Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 20123–20132.
- [18] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. 2014. Microsoft COCO: Common Objects in Context. In *Proceedings of the European Conference on Computer Vision (ECCV) (Lecture Notes in Computer Science)*, Vol. 8693. Springer, 740–755.
- [19] Siyi Liu, Chen Chen, Xiangfeng Du, and Wanli Li. 2024. AgentHPO: Large Language Model Agent for Hyper-Parameter Optimization. *arXiv preprint arXiv:2402.01881* (2024).
- [20] Yury A. Malkov and Dmitry A. Yashunin. 2020. Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42, 4 (2020), 824–836.
- [21] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. *Introduction to Information Retrieval*. Cambridge University Press.
- [22] Shan Ning, Longtian Qiu, Yongfei Liu, and Xuming He. 2023. HOICLIP: Efficient Knowledge Transfer for HOI Detection with Vision-Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 23507–23517.
- [23] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Remi Bourdoukan. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research* (2024).
- [24] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*. PMLR, 8748–8763.
- [25] Harsha Vardhan Simhadri, George Williams, Martin Aumüller, Matthijs Douze, Artem Babenko, Dmitry Postler, Wei Wang, Micha Bern, Jingtao Gao, Peng Zhao, Cecilia Aguerrebere, et al. 2022. Results of the NeurIPS’21 Challenge on Billion-Scale Approximate Nearest Neighbor Search. In *Proceedings of the NeurIPS 2021 Competitions and Demonstrations Track (Proceedings of Machine Learning Research)*, Vol. 176. PMLR, 177–189.
- [26] Jasper Snoek, Hugo Larochelle, and Ryan P. Adams. 2012. Practical Bayesian Optimization of Machine Learning Algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 25.
- [27] Suhas Jayaram Subramanya, Fnu Devvrit, Harsha Vardhan Simhadri, Ravishankar Krishnaswamy, and Rohan Kadekodi. 2019. DiskANN: Fast Accurate Billion-point Nearest Neighbor Search on a Single Node. In *Advances in Neural Information Processing Systems (NeurIPS)*. 13748–13758.
- [28] Dana Van Aken, Andrew Pavlo, Geoffrey J. Gordon, and Bohan Zhang. 2017. Automatic Database Management System Tuning Through Large-scale Machine Learning. In *Proceedings of the 2017 ACM International Conference on Management of Data (SIGMOD)*. 1009–1024.
- [29] Vishakha Vij, Luis Remis, Abhishek Gupta, and Sharif Sakr. 2023. VDMS: An Efficient Big-Visual-Data Access for Machine Learning Workloads. *Proceedings of the VLDB Endowment* 16, 12 (2023), 4058–4061.
- [30] Jianguo Wang, Xiaomeng Yi, Rentong Guo, Hai Jin, Peng Xu, Shengjun Li, Xianguo Wang, Xiangzhou Guo, Chengming Li, Xiaohai Xu, Kun Yu, Yuxing Yuan, Yinghao Zou, Ji Long, Yihua Cai, Zhifeng Li, Zheng Zhang, Yihua Mo, Jun Gu, Ruiyi Jiang, Yi Wei, and Charles Xie. 2021. Milvus: A Purpose-Built Vector Data Management System. In *Proceedings of the 2021 International Conference on Management of Data (SIGMOD)*. 2614–2627.
- [31] Cong Wei, Yang Chen, Haonan Chen, Hexiang Hu, Ge Zhang, Jie Fu, Alan Ritter, and Wenhui Chen. 2024. UniIR: Training and Benchmarking Universal Multimodal Information Retrievers. In *Proceedings of the European Conference on Computer Vision (ECCV)*.
- [32] Tobias Weyand, Andre Araujo, Bingyi Cao, and Jack Sim. 2020. Google Landmarks Dataset v2 — A Large-Scale Benchmark for Instance-Level Recognition and Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2575–2584.
- [33] Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. 2024. Large Language Models as Optimizers. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.
- [34] Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. 2014. From Image Descriptions to Visual Denotations: New Similarity Metrics for Semantic Inference over Event Descriptions. *Transactions of the Association for Computational Linguistics* 2 (2014), 67–78.
- [35] Aixi Zhang, Yue Liao, Si Liu, Miao Lu, Yongliang Wang, Chen Gao, and Xiaobo Li. 2021. Mining the Benefits of Two-stage and One-stage HOI Detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 34. 18209–18220.
- [36] Michael R. Zhang, Nishkrit Desai, Juhan Bae, Jonathan Lorraine, and Jimmy Ba. 2023. Using Large Language Models for Hyperparameter Optimization. *arXiv preprint arXiv:2312.04528* (2023).